



Universidad Nacional Autónoma de México
Facultad de Ciencias

Fundamentos de Bases de Datos

Práctica 1 - Bitácora Ledesma

Alumno:
Ledesma Cuevas Eduardo - 321249050

Profesor:
Gerardo Avilés Rosas

7 de septiembre de 2026

Bitácora.

Sistema operativo y versión.

Procesador: Intel i7-1355U (12) @ 5.000GHz
Memoria: 16 GB
Sistema Operativo: Linux
GNOME DE: 50.4
Kernel: 7.2.3-arch1-2
Arquitectura: x86_64

Distribución.

Arch Linux (rolling)

Versión de la instalación.

Docker version 29.7.2, build a7dcaa6fdb
postgres (PostgreSQL) 18.6 (Debian 18.6-1.pgdg13+2)

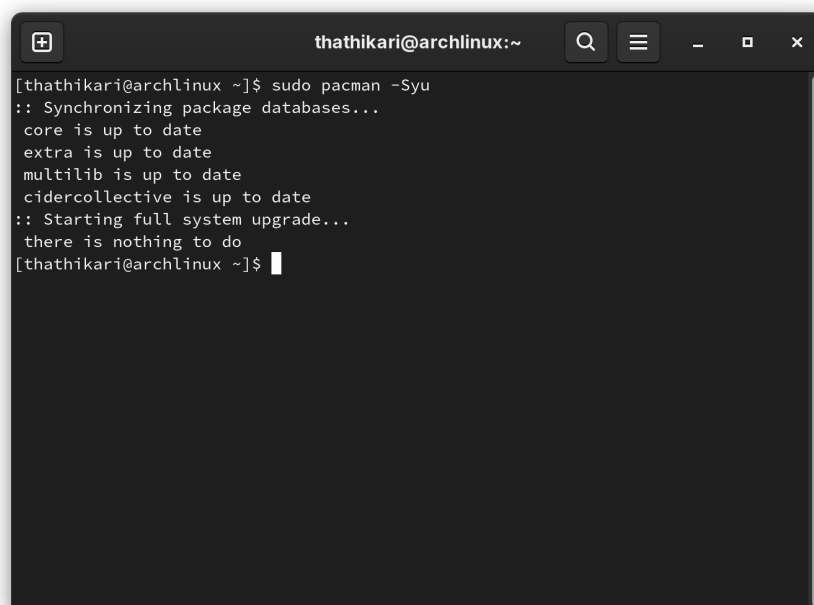
Tiempo requerido.

20 minutos.

Explicación paso a paso.

1. Actualizar el sistema

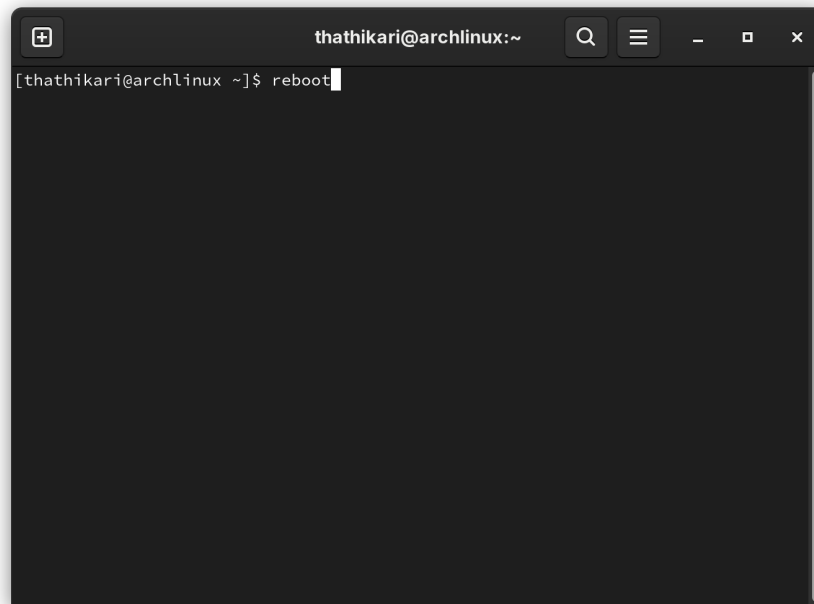
Actualizamos el sistema para asegurarnos de tener las últimas versiones de cada paquete.

A terminal window titled 'thathikari@archlinux:~' with standard window controls. The terminal shows the execution of 'sudo pacman -Syu'. The output indicates that several packages (core, extra, multilib, cidercollective) are up to date, and then a full system upgrade is started, resulting in 'there is nothing to do'.

```
[thathikari@archlinux ~]$ sudo pacman -Syu
:: Synchronizing package databases...
core is up to date
extra is up to date
multilib is up to date
cidercollective is up to date
:: Starting full system upgrade...
there is nothing to do
[thathikari@archlinux ~]$
```

2. Reiniciar el sistema

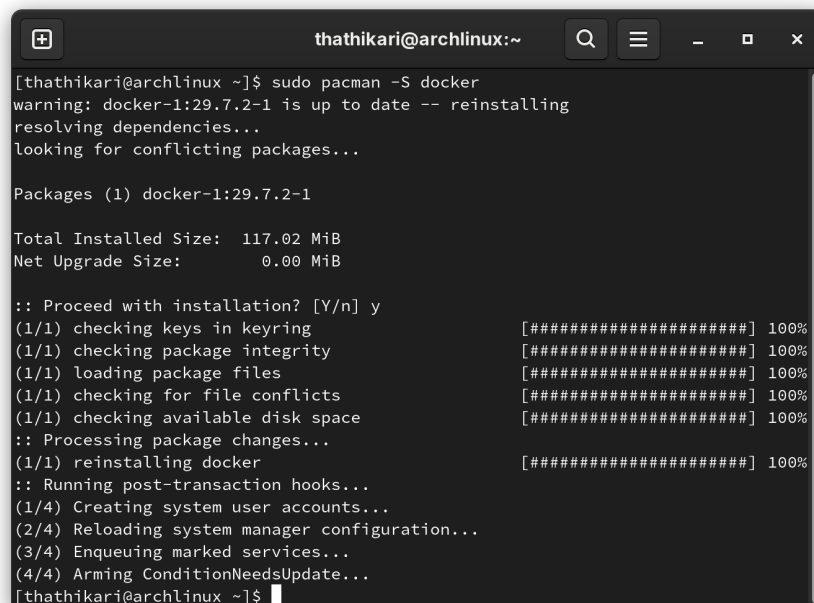
Se recomienda reiniciar el sistema después de una actualización, en caso que haya actualizaciones de *kernel*.



```
thathikari@archlinux:~  
[thathikari@archlinux ~]$ reboot
```

3. Instalar Docker

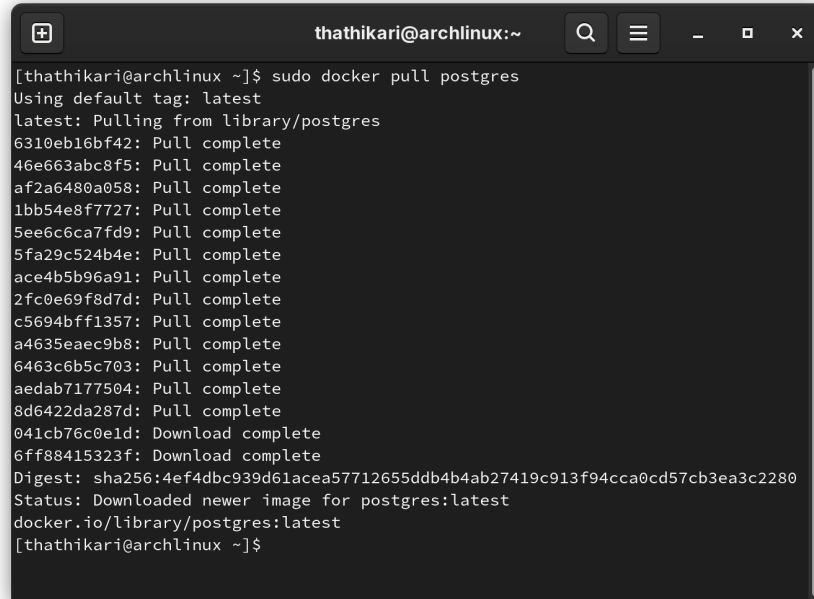
Instalamos Docker, en mi caso el repo de la distribución ya cuenta con el paquete correspondiente.



```
thathikari@archlinux:~  
[thathikari@archlinux ~]$ sudo pacman -S docker  
warning: docker-1:29.7.2-1 is up to date -- reinstalling  
resolving dependencies...  
looking for conflicting packages...  
  
Packages (1) docker-1:29.7.2-1  
  
Total Installed Size: 117.02 MiB  
Net Upgrade Size: 0.00 MiB  
  
:: Proceed with installation? [Y/n] y  
(1/1) checking keys in keyring [#####] 100%  
(1/1) checking package integrity [#####] 100%  
(1/1) loading package files [#####] 100%  
(1/1) checking for file conflicts [#####] 100%  
(1/1) checking available disk space [#####] 100%  
:: Processing package changes...  
(1/1) reinstalling docker [#####] 100%  
:: Running post-transaction hooks...  
(1/4) Creating system user accounts...  
(2/4) Reloading system manager configuration...  
(3/4) Enqueueing marked services...  
(4/4) Arming ConditionNeedsUpdate...  
[thathikari@archlinux ~]$
```

4. Descargar la imagen de postgres

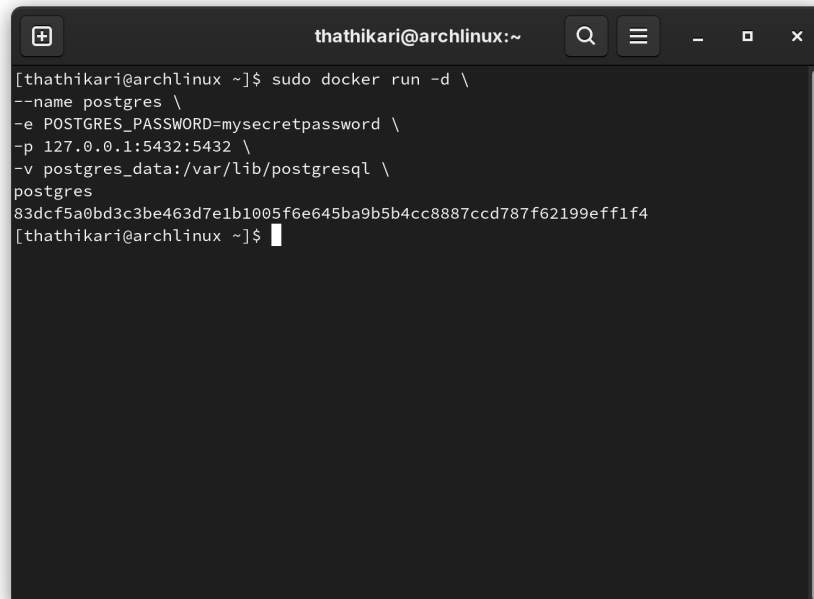
Descargamos la imagen de postgres con `docker pull`.



```
thathikari@archlinux:~  
[thathikari@archlinux ~]$ sudo docker pull postgres  
Using default tag: latest  
latest: Pulling from library/postgres  
6310eb16bf42: Pull complete  
46e663abc8f5: Pull complete  
af2a6480a058: Pull complete  
1bb54e8f7727: Pull complete  
5ee6c6ca7fd9: Pull complete  
5fa29c524b4e: Pull complete  
ace4b5b96a91: Pull complete  
2fc0e69f8d7d: Pull complete  
c5694bff1357: Pull complete  
a4635eaec9b8: Pull complete  
6463c6b5c703: Pull complete  
aedab7177504: Pull complete  
8d6422da287d: Pull complete  
041cb76c0e1d: Download complete  
6ff88415323f: Download complete  
Digest: sha256:4ef4dbc939d61acea57712655ddb4b4ab27419c913f94cca0cd57cb3ea3c2280  
Status: Downloaded newer image for postgres:latest  
docker.io/library/postgres:latest  
[thathikari@archlinux ~]$
```

5. Crear el contenedor

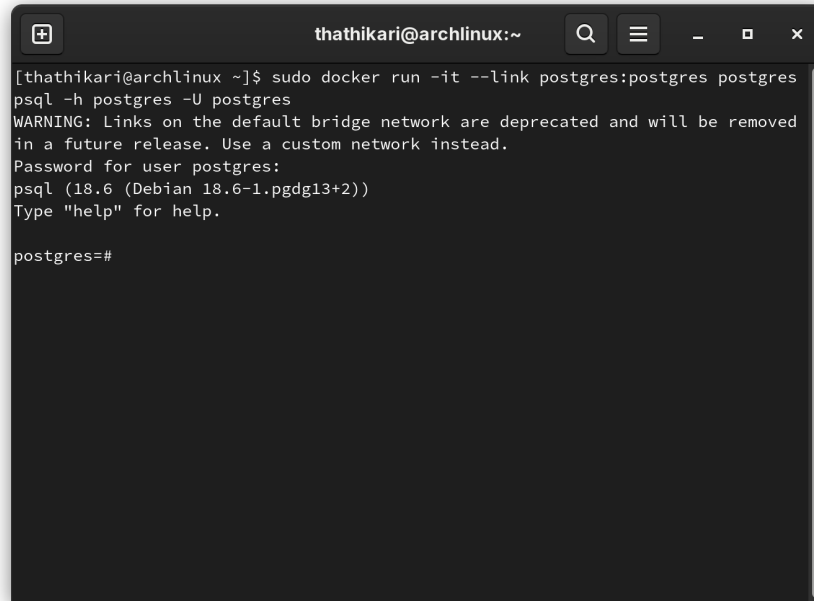
Ejecutamos `docker run` con las opciones de configuración correspondientes.



```
thathikari@archlinux:~  
[thathikari@archlinux ~]$ sudo docker run -d \  
--name postgres \  
-e POSTGRES_PASSWORD=mysecretpassword \  
-p 127.0.0.1:5432:5432 \  
-v postgres_data:/var/lib/postgresql \  
postgres  
83dcf5a0bd3c3be463d7e1b1005f6e645ba9b5b4cc8887ccd787f62199eff1f4  
[thathikari@archlinux ~]$
```

6. Conectarse a la BD a través de psql

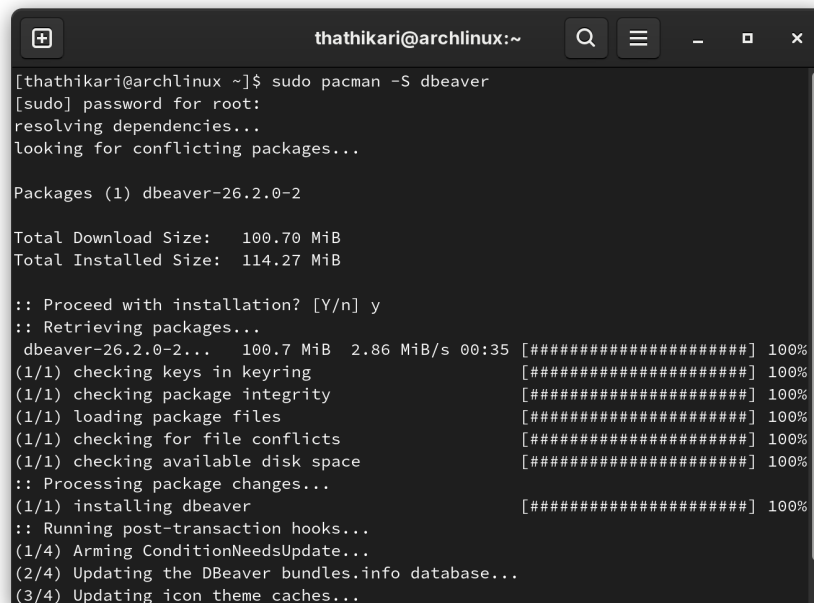
Podemos ejecutar psql dentro del contenedor para conectarnos a la BD.



```
thathikari@archlinux:~  
[thathikari@archlinux ~]$ sudo docker run -it --link postgres:postgres postgres  
psql -h postgres -U postgres  
WARNING: Links on the default bridge network are deprecated and will be removed  
in a future release. Use a custom network instead.  
Password for user postgres:  
psql (18.6 (Debian 18.6-1.pgdg13+2))  
Type "help" for help.  
  
postgres=#
```

7. Instalar DBeaver

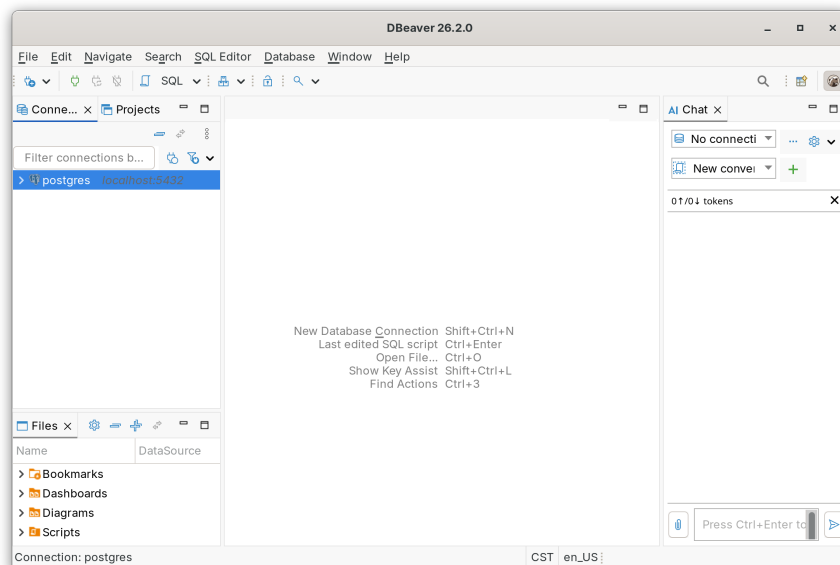
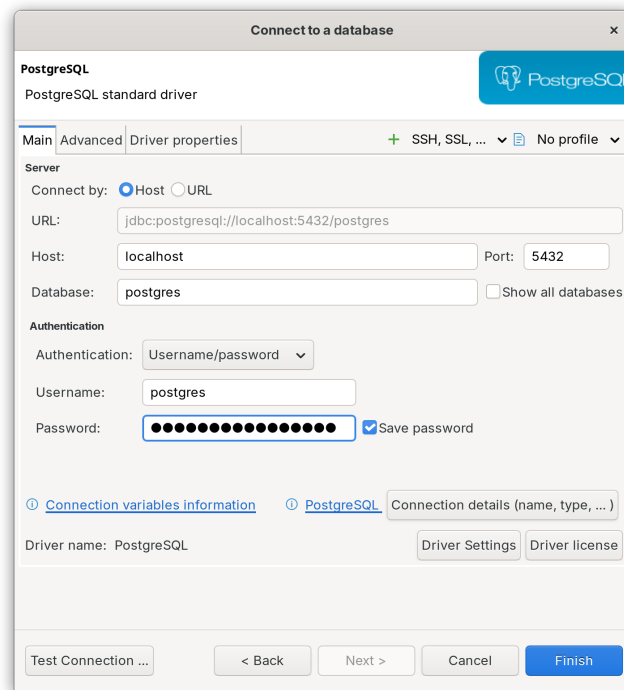
En caso de querer administrar la base de datos con interfaz gráfica, podemos instalar DBeaver.



```
thathikari@archlinux:~  
[thathikari@archlinux ~]$ sudo pacman -S dbeaver  
[sudo] password for root:  
resolving dependencies...  
looking for conflicting packages...  
  
Packages (1) dbeaver-26.2.0-2  
  
Total Download Size: 100.70 MiB  
Total Installed Size: 114.27 MiB  
  
:: Proceed with installation? [Y/n] y  
:: Retrieving packages...  
dbeaver-26.2.0-2... 100.7 MiB 2.86 MiB/s 00:35 [#####] 100%  
(1/1) checking keys in keyring [#####] 100%  
(1/1) checking package integrity [#####] 100%  
(1/1) loading package files [#####] 100%  
(1/1) checking for file conflicts [#####] 100%  
(1/1) checking available disk space [#####] 100%  
:: Processing package changes...  
(1/1) installing dbeaver [#####] 100%  
:: Running post-transaction hooks...  
(1/4) Arming ConditionNeedsUpdate...  
(2/4) Updating the DBeaver bundles.info database...  
(3/4) Updating icon theme caches...
```

8. Conectar DBeaver con PostgreSQL

Conectamos DBeaver con la base de datos usando la contraseña antes configurada.



Comentarios y los problemas a los que te enfrentaste en la instalación.

En general la instalación fue muy sencilla, pues el repositorio de mi distribución ya contaba con los paquetes necesarios. El único problema que encontré fue que después de actualizar el sistema, hubo que reiniciar el sistema para que el daemon de Docker iniciara correctamente.